

Introduction to machine learning with WatsonX

Advanced Practical Course

The next two days

Today

- Introduction to ML
- Exercises
- SAP Experience Center

Tomorrow

- Introduction genAI
- Introduction AI Agents
- Interactive Agent Game

What is Machine Learning?

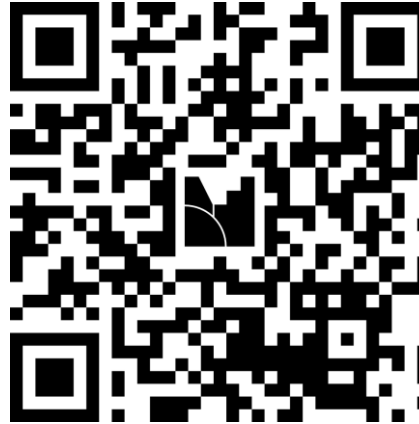
„[Machine learning is the] field of study that gives computers the ability to learn without being explicitly programmed.”

-- Arthur Samuel, 1959

„A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

-- Tom Mitchell, 1997

What is Machine Learning?



What applications of machine learning do you know in everyday life?

What is Machine Learning?

„A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .“

-- Tom Mitchell, 1997

e.g. spam filter



T:

E:

P:

What is Machine Learning?

„A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .“

-- Tom Mitchell, 1997

e.g. spam filter

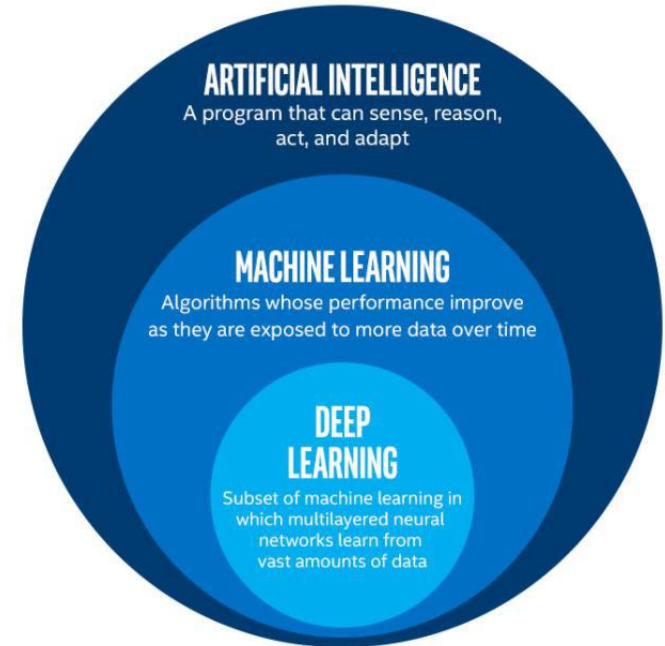
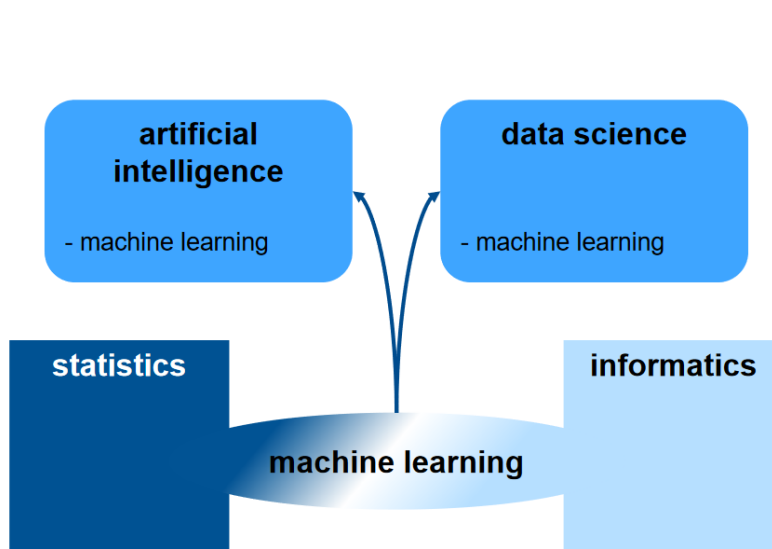


T : flag spam for new mails

E : training data of flagged mails

P : ratio of correctly classified emails = accuracy

What is Machine Learning?



...and a lot of other areas

Contents

- Why Machine Learning?
- CRISP-DM Cycle
- Machine Learning Categories
- WatsonX
- LLMs on WatsonX
- Practical exercise

Why Machine Learning?

Why Machine Learning?

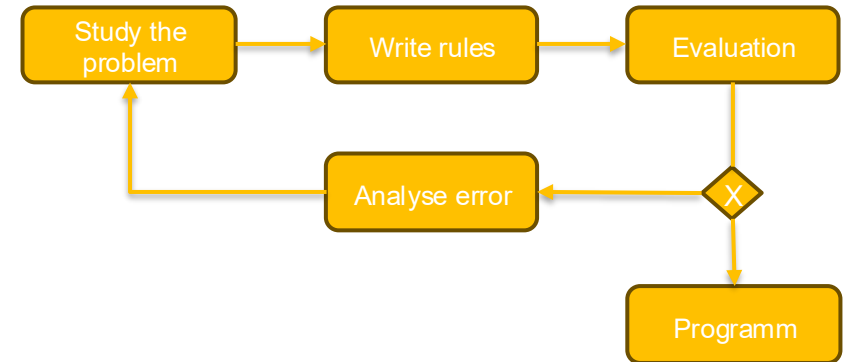
How to write a spam filter?

Discuss for 5-10 minutes how you would approach this problem and briefly write down your approach.

Why Machine Learning?

How to write a spam filter?

- 1) Understand the problem e.g. What does a spam mail normally look like?
- 2) Write all necessary rules (manually)
- 3) Test and repeat



Why Machine Learning?

How to write a spam filter?

Rulebased Example:

[Link](#)

Why Machine Learning?

How to write a spam filter?

Rulebased Example:

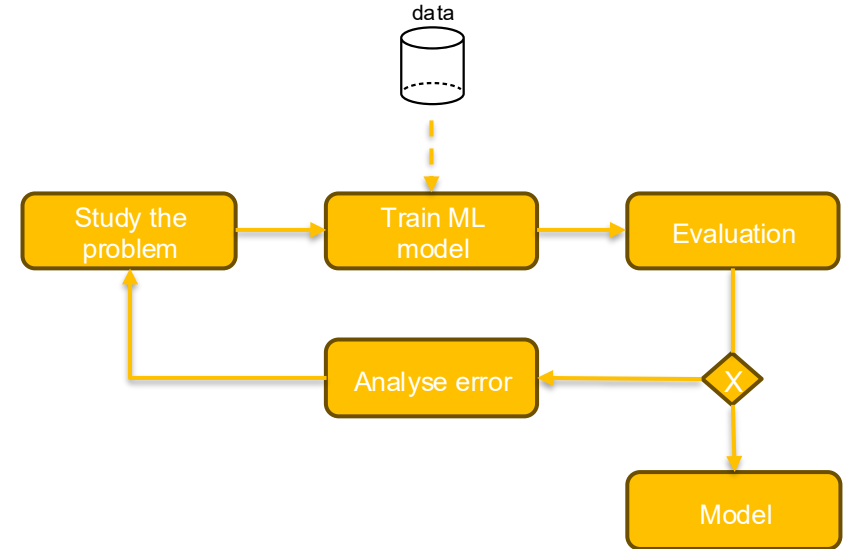
[Link](#)

Discuss for 5 minutes what the disadvantages of this approach are.

Why Machine Learning?

How to write a spam filter with machine learning?

- 1) Understand the problem e.g. What does a spam mail normally look like?
- 2) Train the model
- 3) Test and repeat



Why Machine Learning?

How to write a spam filter with machine learning?

MLbased Example:

[Link](#)

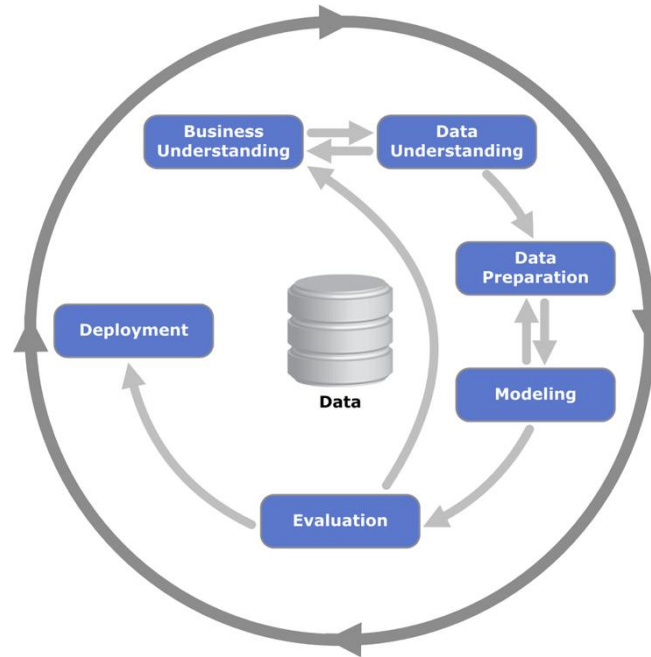
Discuss for 5 minutes what the (dis-)advantages of this approach are.

CRISP-DM Cycle

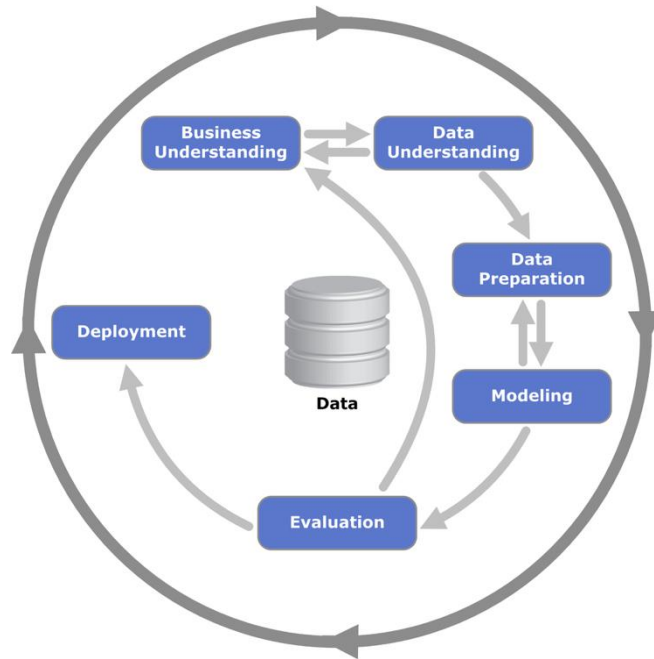
CRISP-DM process model

- **C**Ross-Industry **S**tandard **P**rocess for **D**ata **M**ining – CRISP-DM
 - As a methodology, it includes descriptions of the typical phases of a project, the tasks involved with each phase, and an explanation of the relationships between these tasks
 - As a process model, CRISP-DM provides an overview of the data mining life cycle
 - Advantages: Standardization, shorter project runtimes, cost reduction, reusability

CRISP-DM process model



CRISP-DM process model

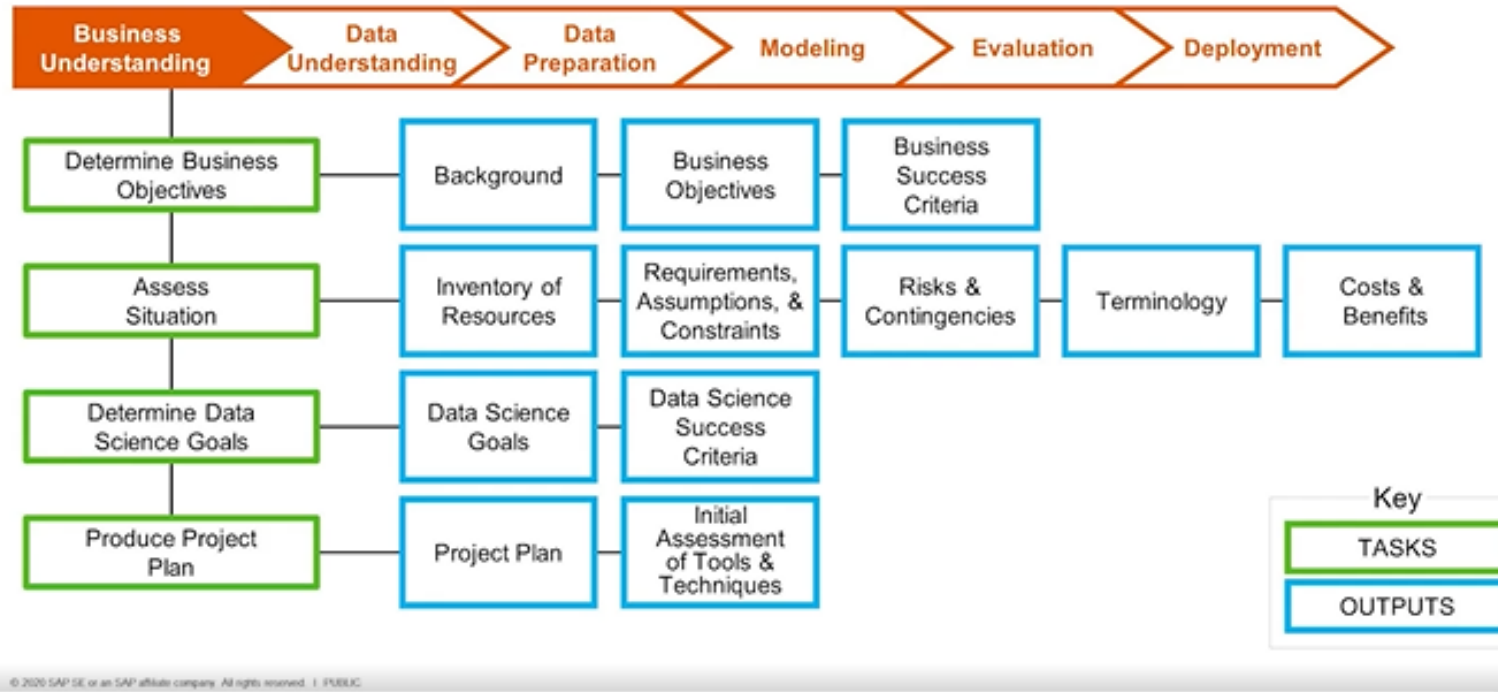


What is the most important phase in the CRISP-DM process model?

Business Understanding

- Before applying any data mining technologies, it is necessary to understand what you want to achieve
- Talk with experts to get an understanding of the domain
- Develop an understanding of the project objectives or the business problem
- Decision whether data mining can solve the problem and selection of one or more methods
- Determine what kind of data is needed and identify available data sets in the domain
- Draft of an initial project plan to achieve the project objectives

Business Understanding

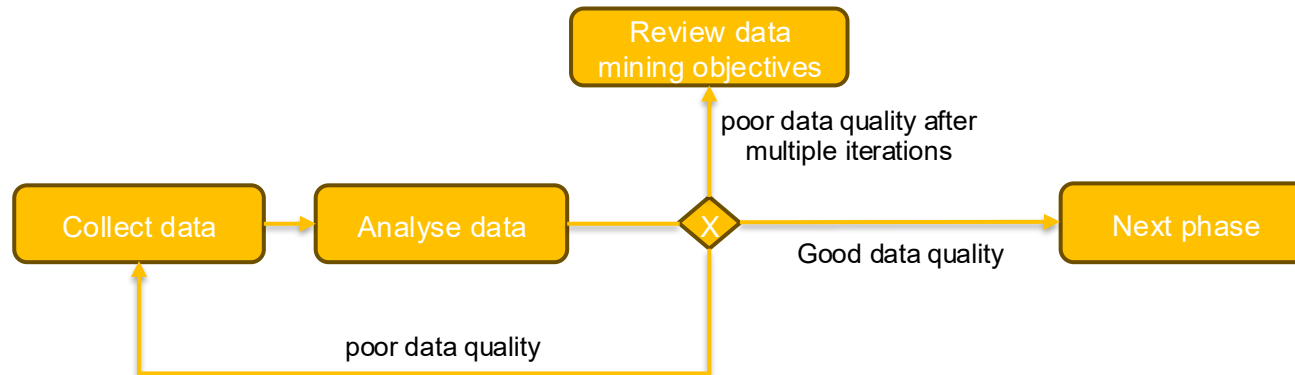


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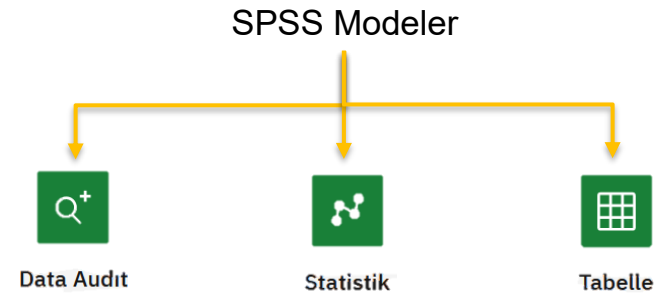
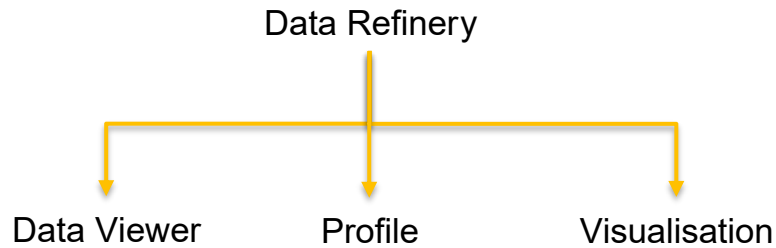
Illustration of SAP SE

Data understanding

- Collecting data
- Understanding the data
- First insights into the data
- Identifying data quality issues ☐ Use data statistic tools



Data understanding with watsonx.ai



Data understanding with Data Audit



Data Audit

Gives a first interactive overview of the dataset with the following information:

- Information such as evaluation statistics, histograms, box plots, bar charts, pie charts, etc., which can be useful for a preliminary understanding of the data
- Information on outliers, extreme values and missing values
- Further information: [WatsonX Documentation](#)

Data understanding with Data Audit



Iris Dataset (Pedregosa et al. 2011, Dua/Graff 2017)

Data Audit



Iris Versicolor



Iris Setosa



Iris Virginica

Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; Vanderplas, J.; Passos, A.; Coumapeau, D.; Brucher, M.; Perrot, M.; Duchesnay, E. (2011): Scikit-learn: Machine Learning in Python. In: Journal of Machine Learning Research, Vol. 12 (2011), pp. 2825–2830.

Dua, D.; Graff, C. (2017): UCI Machine Learning Repository. University of California, Irvine, School of Information and Computer Sciences.

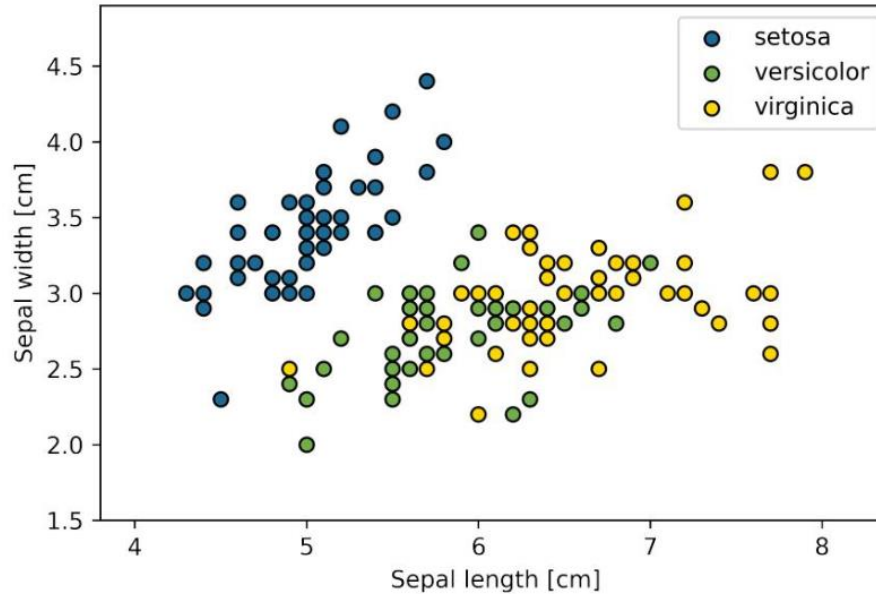
<https://www.kaggle.com/datasets/vikrishnan/iris-dataset>

Data understanding with Data Audit



Iris Dataset

Data Audit

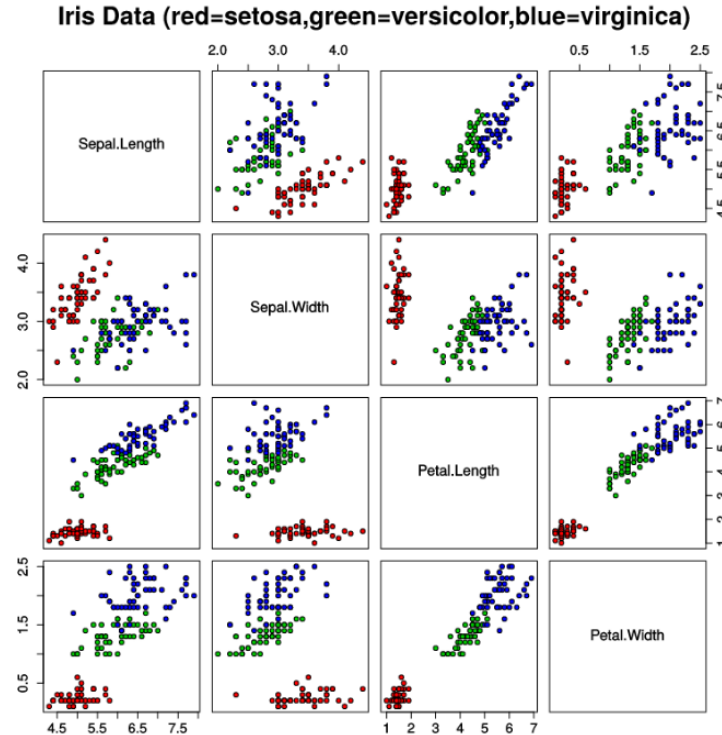


Data understanding with Data Audit



Data Audit

Iris Dataset



Data understanding with Data Audit



Iris Dataset - Overview

Data Audit

Name	Diagramm	Messwert	Typ	Min	Max	Mittelwert	Std. Dev
sepal_length		 Fortlaufend	Doppelt	4,300	7,900	5.843	0.828
sepal_width		 Fortlaufend	Doppelt	2,000	4,400	3.054	0.434
petal_length		 Fortlaufend	Doppelt	1,000	6,900	3.759	1.764
petal_width		 Fortlaufend	Doppelt	0,100	2,500	1.199	0.763
species		 Nominal	Zeichenfolge	Iris-setosa	Iris-virginica	--	--

Data understanding with Data Audit



Iris Dataset – Data Quality

Data Audit

Name	Messwert	Typ	Ausreißer	↑↓	Extremwerte	% abgeschlossen	Gültig
sepal_length	✎ Fortlaufend	Doppelt	0		0	100%	150
sepal_width	✎ Fortlaufend	Doppelt	4		0	100%	150
petal_length	✎ Fortlaufend	Doppelt	0		0	100%	150
petal_width	✎ Fortlaufend	Doppelt	0		0	100%	150
species	☼ Nominal	Zeichenfolge	--		--	100%	150

Data understanding with Data Audit



Data Audit

petal_length

Statistiken Analyse

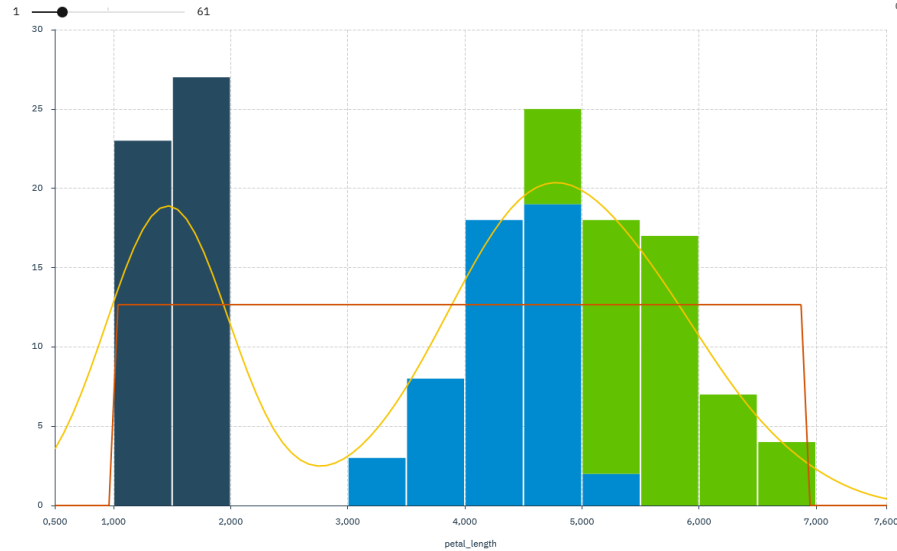
Letztes Profil: 04-Jul-2024 15:12 150 Zeilen

Histogramm

Boxplot

Q-Q-Kurven

Verteilung



Basisstatistiken ①

Minimum 1	Maximum 6.90
Gemittelt 4.35	Durchschnitt 3.76
Summe 563.80	Modus 1.50
Gültig 150	Kardinalität 0.29
Eindeutig 10	Spanne 5.90
Einzelne Werte 43	Fehlende Werte 0
Ausreißer 0	

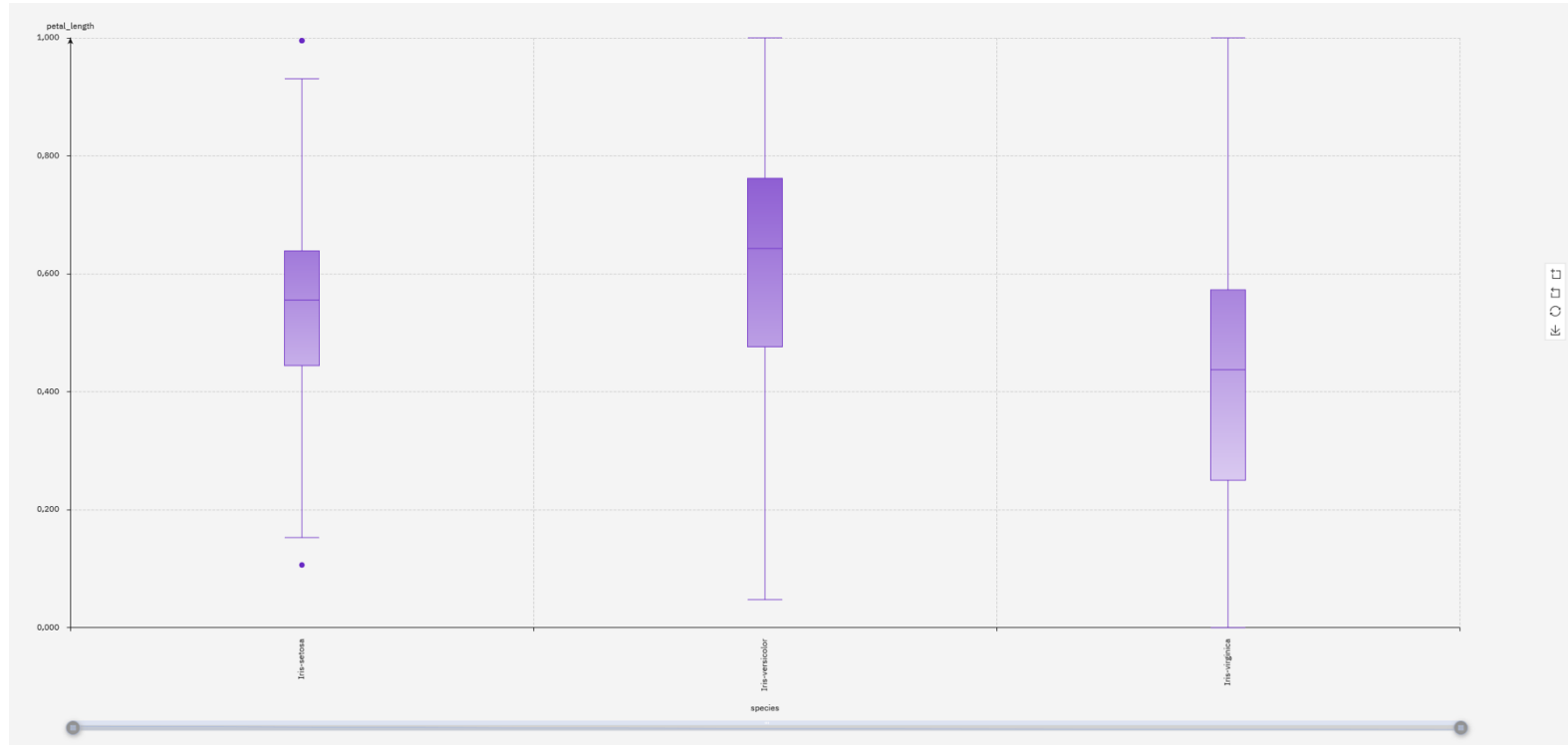
Erweiterte Einblicke ①

25. Perzentil 1.60	75. Perzentil 5.10
Std. Abweichung 1.76	Variance 3.11

Data understanding with Data Audit



Diagramme



Data understanding with Data Audit



Iris Dataset

Statistik

- Statistical key figures for individual numerical characteristics
- Checking the (Pearson) correlation between two numerical characteristics

⊖ sepal_length

⊖ Statistik

Anzahl	150
Mittelwert	5.843
Min	4.300
Max	7.900
Bereich	3.600
Varianz	0.686
Standardabweichung	0.828
Standardfehler des Mittelwerts	0.068
Median	5.800

Data understanding

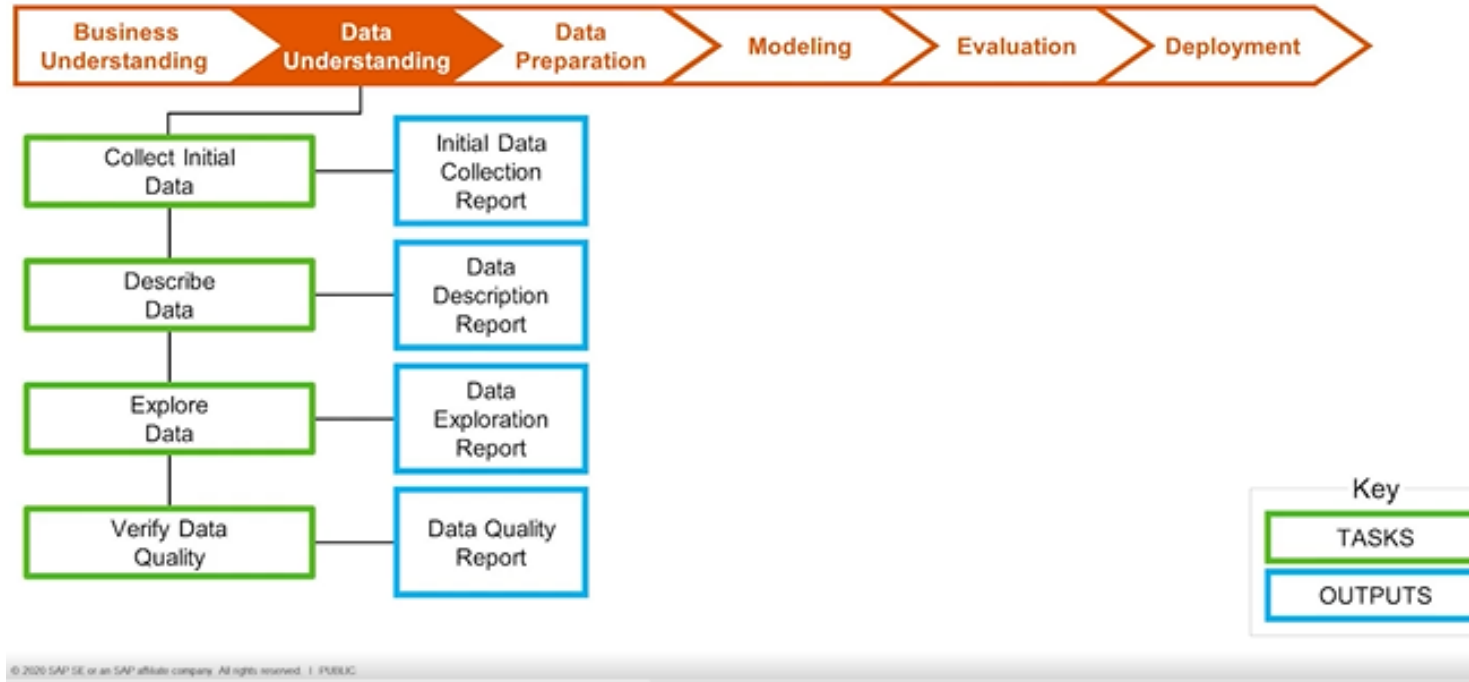


Illustration of SAP SE

Data preparation

- All activities necessary to create a data set for machine learning
- Iterative process:



- Activities:
 - Table, data set and feature selection (feature selection)
 - Data cleansing considering the modelling tool and the selected algorithm
 - Transformation of the data considering the modelling tool and the selected algorithm
 - Data reduction

Data preparation

How can you deal with missing data?

Data preparation – Missing data

- Ignore the data records
- Manual retrospective capture of missing values
- Replace missing values with global constants
- Replace missing values with the average of all values
- Replace missing values with the mean value of its values in this class
- Replace missing values with the most probable value → e.g. through regression

Data preparation – Inconsistencies

- Correction of inconsistencies through manual correction
- Inconsistencies due to problems in data integration e.g. different characteristics with the same name and different meanings (homonyms) must be renamed
- Feature Selection: Remove redundancies that occurred due to data integration

Data preparation

1994 Census bureau database

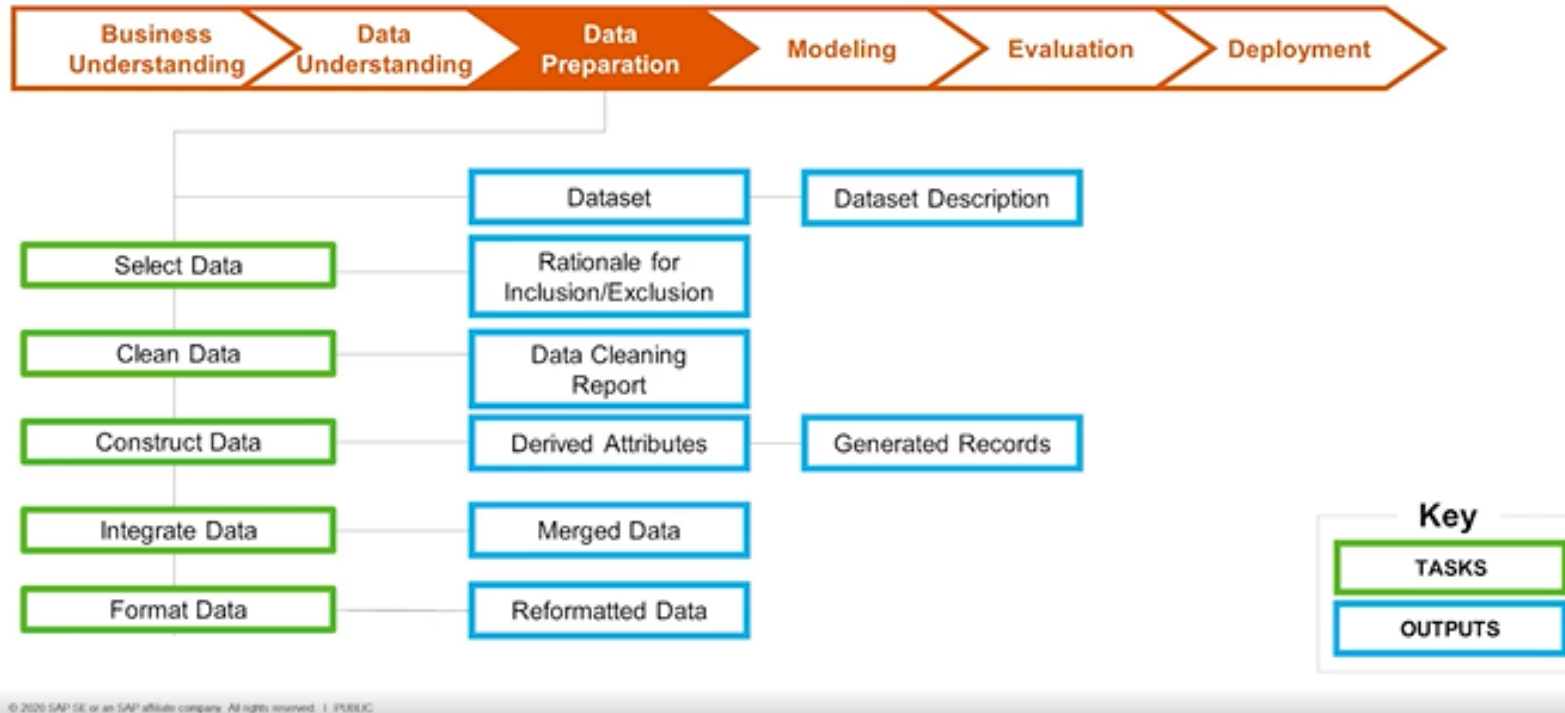
- 15 Characteristics
- US-Americans between 18 and 90 years of age
- General characteristics
- Education, work, relationship
- Annual salary

Task: Form groups of 4 people; each group should analyse the data set regarding data issues and find ways to solve them. Prepare a short presentation.

by Ronny Kohavi and Barry Becker (Data Mining and Visualization, Silicon Graphics)

<https://www.kaggle.com/datasets/uciml/adult-census-income/>

Data preparation



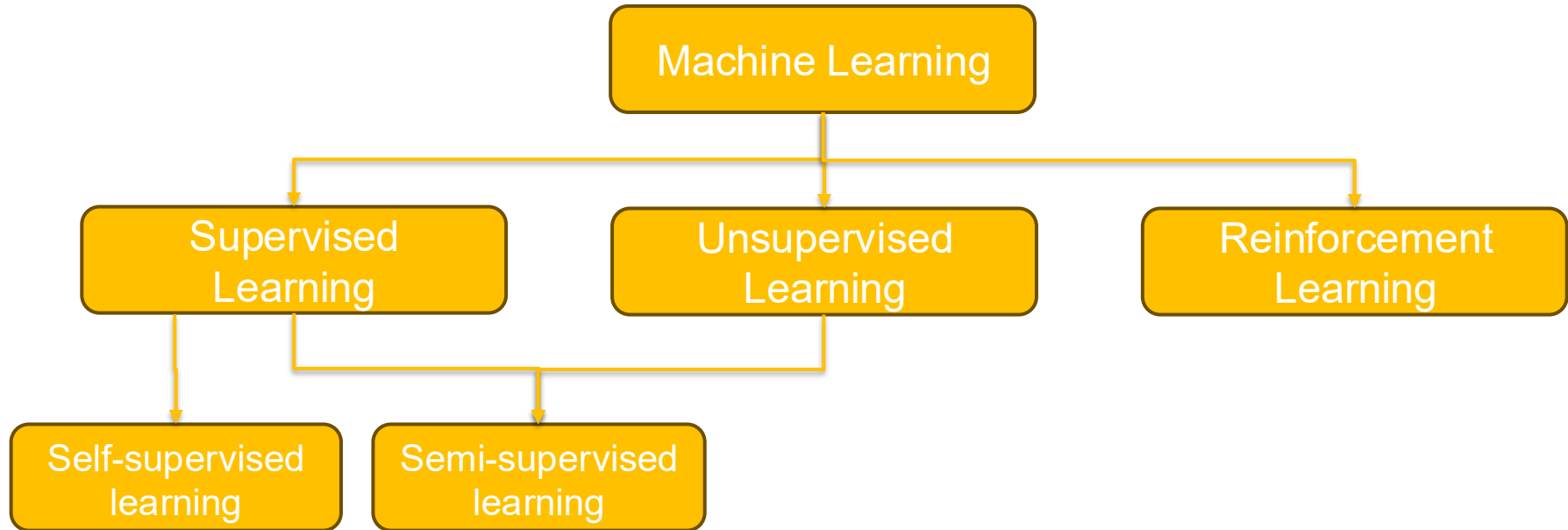
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Illustration of SAP SE

Modeling

- Selection and application of statistical methods and machine learning methods for the development of models
- Deciding which ML approach is the best one for the data
- Setting up a system environment for developing/training the models
- Preparation of the data according to the requirements of the algorithm

Modeling - ML approaches



More details in chapter „Machine Learning Categories”

Modeling

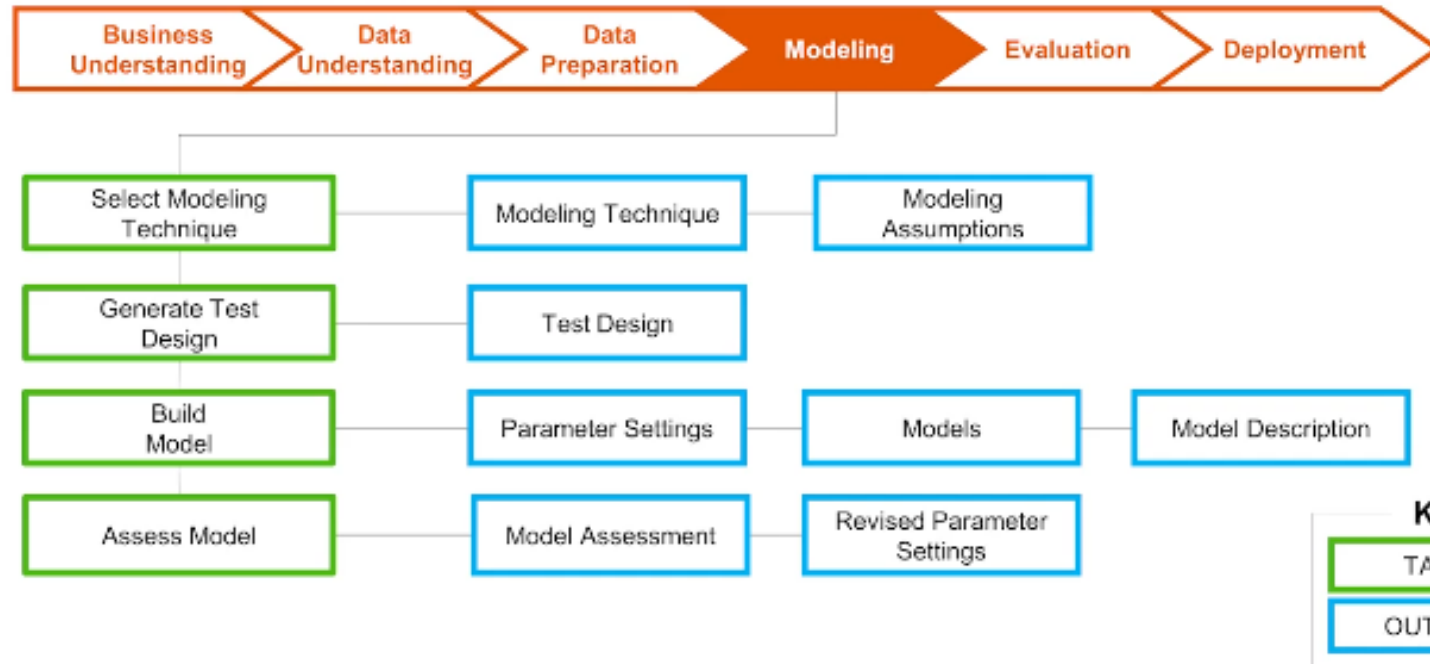


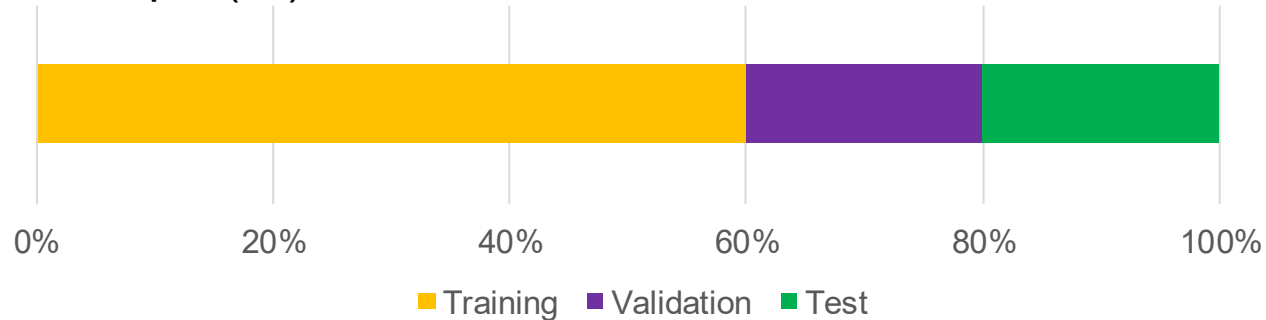
Illustration of SAP SE

Evaluation

- Once a model has been successfully generated, evaluation is a key point
- Without evaluation, there is a risk that a model will generate incorrect results, results that cannot be generalised or results that do not fit the business purpose
- A decision on the model can only be made after the evaluation

Evaluation

- In order to train and evaluate a model, the available data must be divided up accordingly
- Two example options:
 - Train Test Split (fix)

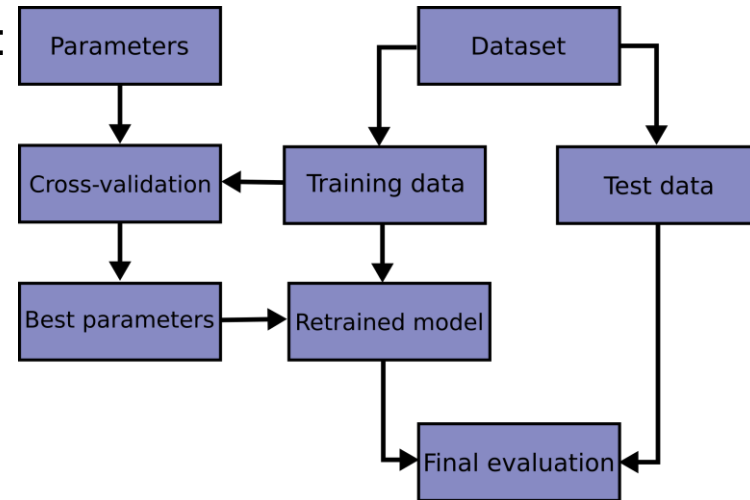


- Cross-validation

Evaluation

Cross-validation (k-fold)

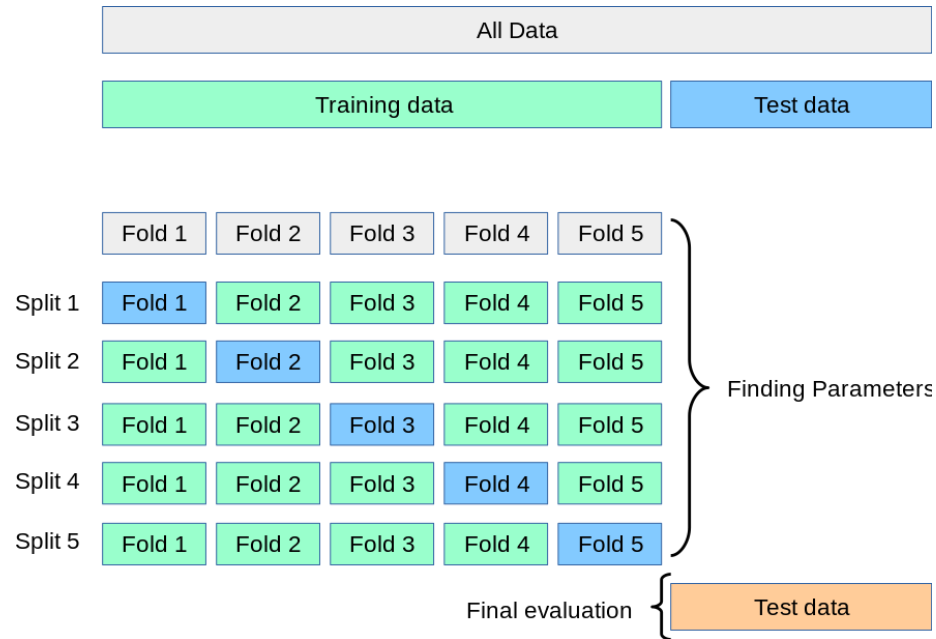
- Training set is split into k smaller sets
- For each k set the following procedure applies:
 - A model is trained using of the folds as training data
 - the resulting model is validated on the remaining part of the data (i.e., it is used as a test set to compute a performance measure such as accuracy)
- Main advantage is reduced data waste, because no fixed validation set is used, but it is computationally expensive



https://scikit-learn.org/stable/_images/grid_search_workflow.png

Evaluation

Cross-validation (k-fold)



https://scikit-learn.org/stable/_images/grid_search_cross_validation.png

Evaluation

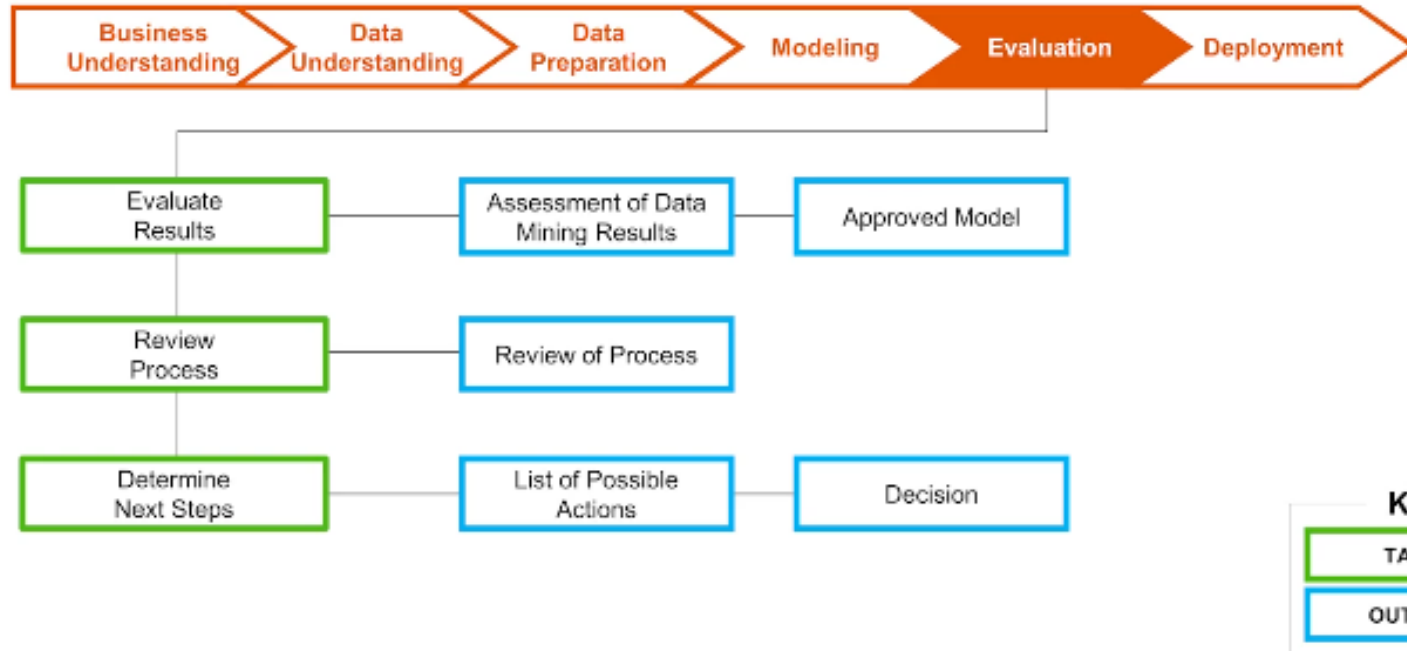


Illustration of SAP SE

Deployment

- The model must be deployed to be available for the use case
- For example, WatsonX offers three deployment modes
 - Online Deployment: process input data in real-time
 - Batch Deployment: process a large batch of input data from a data source and write the output to a selected destination
 - Application deployment: Create an application deployment to deploy your application assets, such as R Shiny applications

Deployment

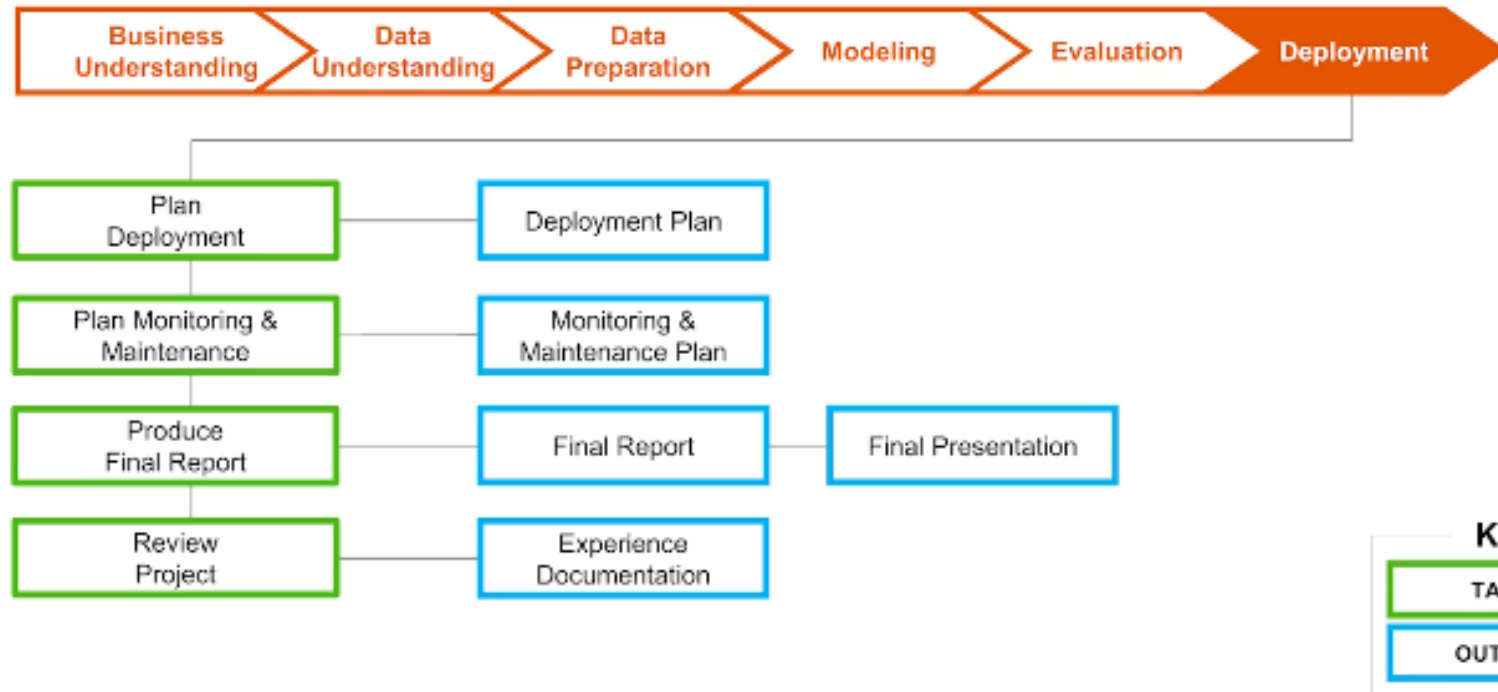
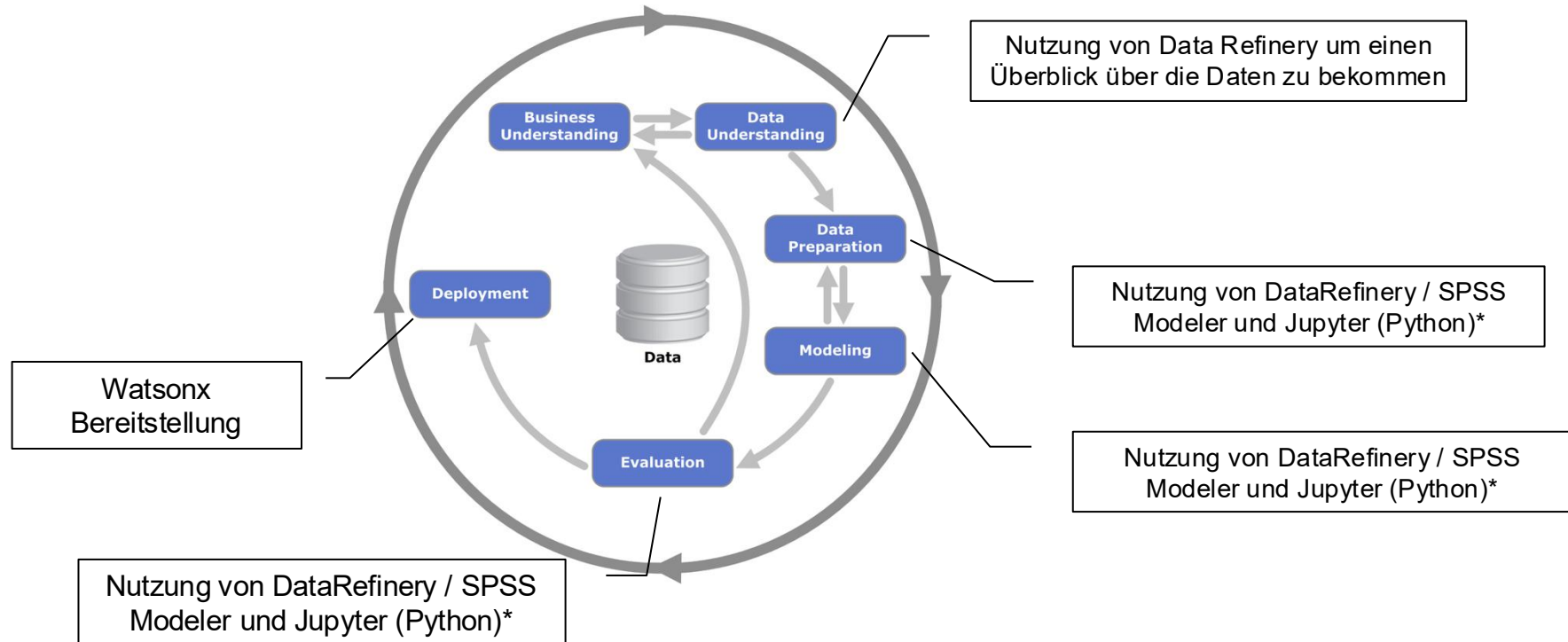


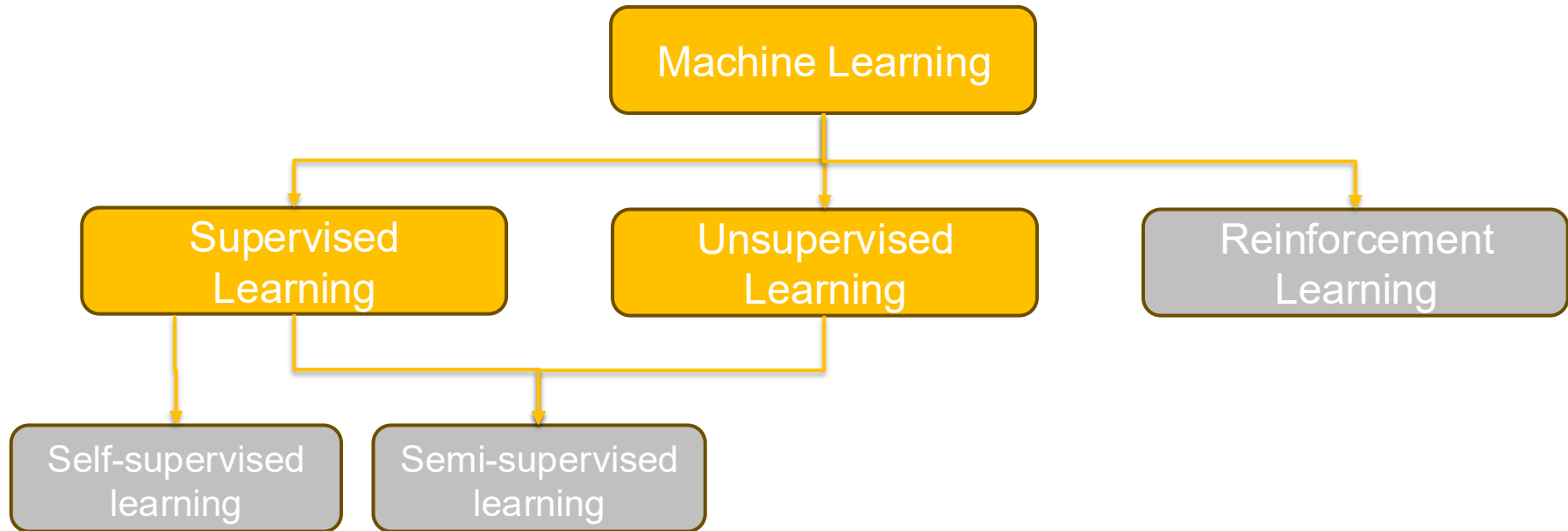
Illustration of SAP SE



(*) NoCode, LowCode, ProCode Optionen

Machine Learning Categories

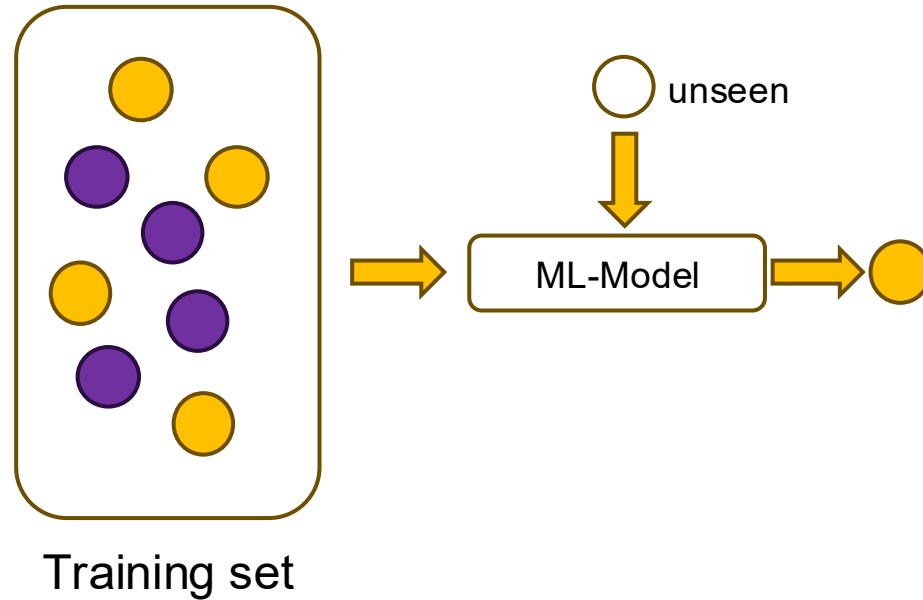
Overview



Supervised Learning

- Use of labeled data sets to train algorithms that to classify data or predict outcomes accurately
- Input: Training dataset with input variables and assigned outputs
- Usable for two types of problems:
 - Classification
 - Regression
- Use Case: Spam Filter, Sales Forecast, Image recognition, Sentiment analysis

Supervised Learning



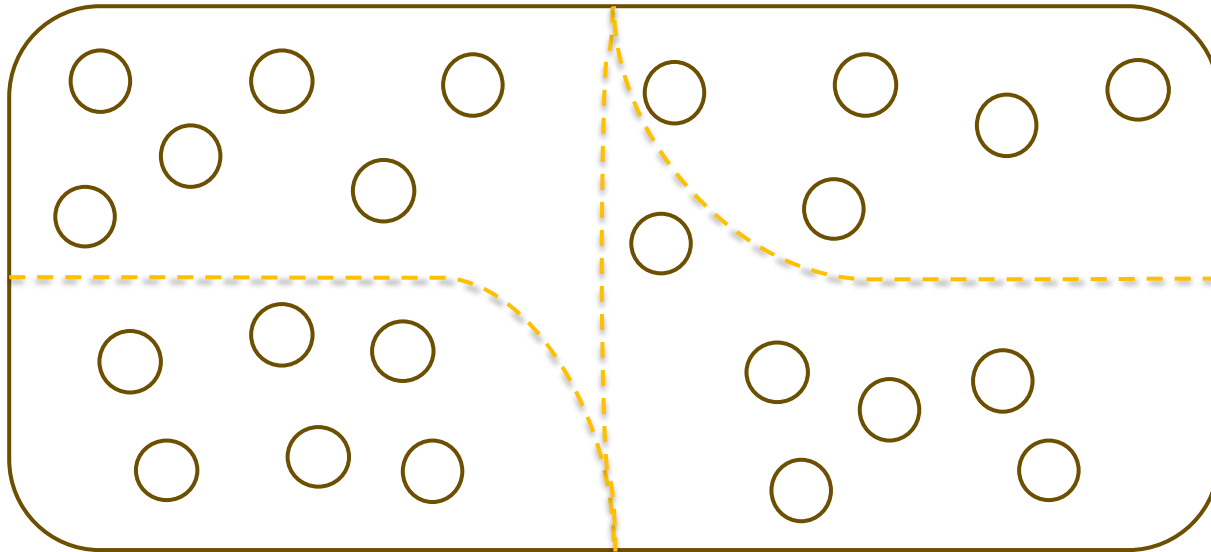
Supervised Learning - Algorithms

- Neural networks
- Naive bayes
- Linear regression
- Logistic regression
- Support vector machines
- Decision trees
- Random forest

Unsupervised Learning

- Algorithms to analyze and cluster unlabeled data sets and discover hidden patterns or data groupings without the need for human intervention
- Input: Training dataset with input variables and without assigned outputs
- Usable for three types of problems:
 - Clustering
 - Association
 - Dimensionality reduction
- Use Case: Anomaly detection, Recommendation engines

Unsupervised Learning



Unsupervised Learning - Algorithms

- K-means clustering
- Hierarchical cluster analysis
- Apriori algorithms
- Principal component analysis (PCA)

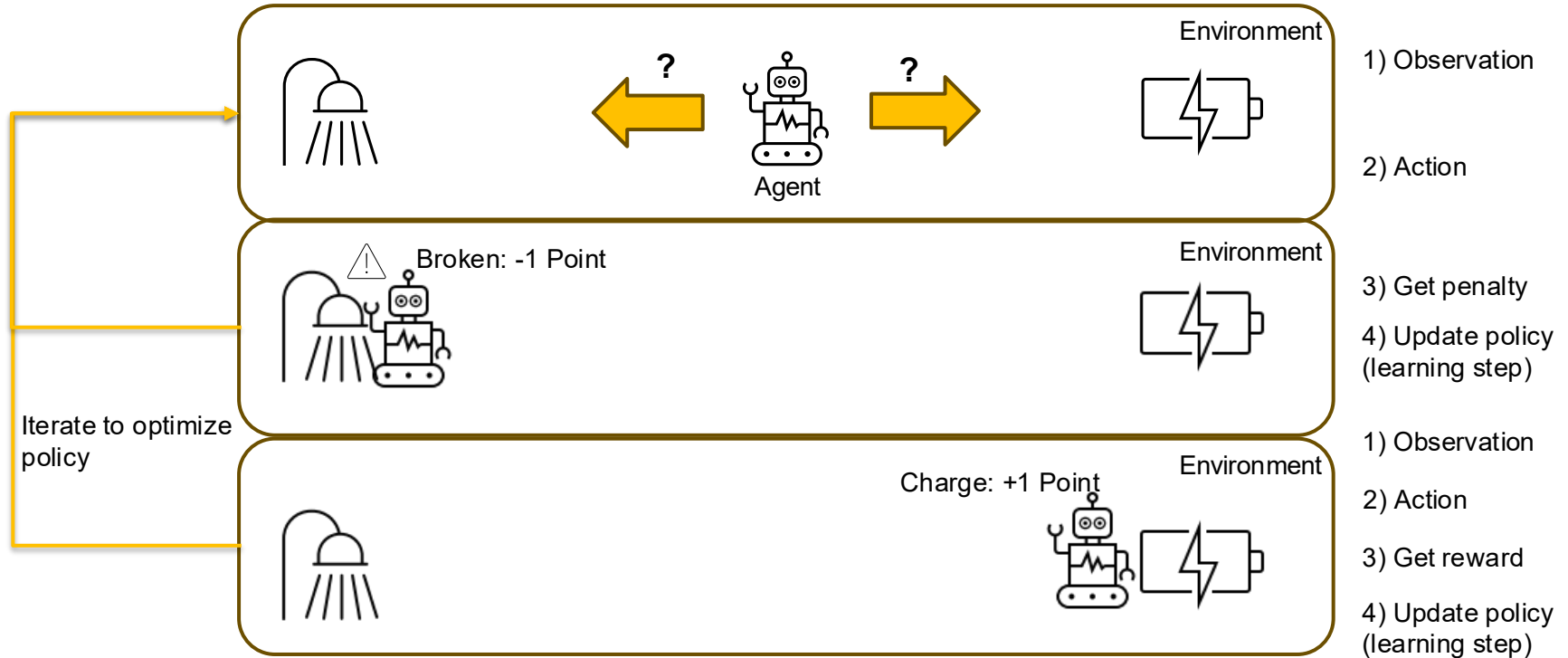
(Un-)Supervised Learning - Demo

WatsonX Demo

Reinforcement Learning

- An agent learns to make decisions by interacting with an environment. It is used in robotics and other decision-making settings.
- Input: Environment with ability to learn by trial-and-error and reward function
- Usable for learning a policy to get the most reward over time
- Use Case: Robotics, Playing games (AlphaGo)

Reinforcement Learning



Reinforcement Learning - Algorithms

- Dynamic programming
 - Monte Carlo
 - Temporal difference learning
-
- Example: <https://youtu.be/qv6UVOQ0F44?feature=shared&t=120>

WatsonX

WatsonX - Overview



watsonx.ai

Train, validate, tune and deploy models for generative AI

[Start building AI applications →](#)



watsonx.governance

Manage, monitor and govern all your AI applications and models

[Govern your AI models →](#)



watsonx.data

Bring all your business data together to scale analytics and AI in your applications

[Manage your data for AI →](#)



watsonx Assistant

Get started with a pre-built application for building and deploying AI customer care assistants

[Pre-built app](#)

[Build your chatbot now →](#)

WatsonX - Overview

Daten vorbereiten ⓘ



Verbindung zu einer Datenquelle herstellen

mit Verbindung



Daten vorbereiten und visualisieren

mit Data Refinery



Wiederverwendbare Parametergruppen
definieren

mit Parametersatz



Synthetische Tabellendaten generieren

mit Synthetic Data Generator

WatsonX - Overview

Mit Modellen arbeiten ⓘ



Foundation Model mit gekennzeichneten Daten optimieren

mit Optimierungsstudio



Mit Basismodellen chatten und Eingabeaufforderungen erstellen

mit Prompt Lab



Modelle für maschinelles Lernen automatisch erstellen

mit AutoAI



Modelle als visuellen Flow erstellen

mit SPSS Modeler



Mit Daten und Modellen in Notebooks mit Python oder R arbeiten

mit Jupyter-Notebookeditor



Optimierungsprobleme lösen

mit Decision Optimization



Modelle mit verteilten Daten trainieren

mit Federated Learning



R-Notebooks und -Script schreiben

mit RStudio

WatsonX - Overview

Modelllebenszyklen automatisieren ⓘ



Modelllebenszyklen automatisieren

mit Pipelines

Demo

WatsonX ML